DEEP FAKE AUDIO DETECTION USING DEEP LEARNING

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***Abstract*—Lately, the advancements in deep learning- based voice synthesis have led to deep fake audios, which are a major security threat, as they can lead to digital identity theft and propagation of false information. The detection of these deep fake audios is difficult because advanced voice cloning models are able to produce a voice like a real human voice. Deep literacy - rooted framework is created in order to find the synthetic speech, this being a major concern nowadays. Basically, a Convolutional Neural Network (CNN) is used in the proposed method to analyze time-frequency representations of the audio signals.**

Aural point maps are generated for each utterance from Mel-scale spectral representations such as MFCC and log-Mel features, which are then fed to the model that learns the fine differences between the genuine and manipulated speech.

The model is exposed to a dataset comprising real and ai-driven samples for training and testing, to evaluate its detection power. It scores an average of 95% accuracy approximately. The outcomes also demonstrate how well the system can classify genuine and deep-fake audio and the proposed method's dependability. Fraud detection, media authentication, and voice recognition are some of the areas, where the proposed system can be utilized to enhance the level of trust in the digital communication platform.

***Index Terms*—Deepfake Audio, MFCC, CNN–LSTM, Deep Learning, Voice Cloning, Spectrogram Analysis.**

I. Introduction

Artificial intelligence has seen very quick progress, especially in the areas of speech conflation and audio generation. Technologies like voice cloning and advanced text-to-speech systems are making speech so realistic that it almost sounds like real human voices. Still, the fact is that these features not only can offer come great opportunities but are also capable of being used for harm. Fake audio clips can easily mislead people or even harm them if nobody verifies them." That said, false audio recordings can make people believe that they are hearing a third person. In this way, a dishonest person can communicate lies, commit frauds, identity theft and cybercrime. Furthermore, the risk of voice manipulation is heightened by the fact that voice-controlled facilities are increasingly being utilized in banks, virtual assistants, and authentication challenges. Hence, the issue of identifying such fake audio is becoming a major concern that experts have to address.

Deepfake audio differs from conventional audio editing in that it is produced by advanced neural networks that can not only imitate the speaker's voice but also their intonation, pitch, and mode of speech perfectly. Whereas changes as simple as cutting or introducing noise are quite far from the synthetic audio, which normally only has a few, very slight traces. These are so small that they can hardly be detected by the human ear alone, which is why automated detection systems are indispensable. As deepfake technology gets into the hands of ordinary people, i. e. those who are not experts, it is possible to create fake speech themselves, which is a reason for the further necessity of solid detection techniques.

Some have Graph spectral features, rather than traditional features, are helpful to the networks as they assist in locating the hidden acoustic defects that are inherently produced by the speech synthesis process. Reverse perturbation and internal representation analysis, which are the major methods of detecting differences that even a human being to the very least areas, are examples of the means through this issue is explored. Besides deep learning methods, traditional machine learning methods should not be overlooked in the detection of deepfake audios. Such as Mel-frequency cepstral coefficients (MFCCs), which are usually combined with different types of classifiers like Support Vector Machines (SVM), Random Forests, and Decision Trees. Since they are very simple and work efficiently, these types of methods are very often used for such situations where also i limited data or computational resources. However, these methods may be unsuccessful to identify very advanced synthetic voices, especially when the model is trained on small or language-specific datasets.

Despite the fact that there has been significant breakthroughs in fake audio detection, this field remains vulnerable mainly due to the constantly evolving audio generation techniques, a lack of large diverse datasets, and the difficulty of working well across different languages and recording conditions. Apart from that, majority of the current systems use relatively high computational power, which limits their practical application. This paper attempts to address these issues by presenting an effective signal deepfake audio system, as well as a well-optimized classifier. Roberts et al. Essentially, the main aim is to strike a balance between efficient detection ability and low computational cost. This method may be advantageous.

II. Related Work

Deepfake audio detection is a common issue researchers are currently facing due to the increase in deepfake audio-related issues, which has attracted researchers to work on it by using technologies like neural speech synthesis and voice conversion technologies. Several studies have examined different machine literacy and deep literacy ways for this task [1],[2],[8].

Todorov et al. delved handcrafted aural features including Mel-frequence Cepstral Portions (MFCCs) and Constant-volume Cepstral Portions (CQCCs) together with Gaussian Mixture Models (GMMs) for caricatured speech discovery. This approach is manual and has very limited robustness when tested on unseen spoofing methods[3]. Similarly, Lavrentyeva et al. (2019) applied Support Vector Machines (SVMs) with spectral features; however, the model showed limited performance on unseen data after training on a specific dataset, which remained a challenge [8].

With the advancement of deep learning, researchers had moved forward with convolution-based architectures. Zhang et al. (2020) had implemented Convolutional Neural Networks (CNNs) on log-Mel spectrogram representations of audio signals [7]. Their results handed practical perceptivity and captured spectral characteristics produced by textbook-to-speech (TTS) and voice conversion (VC) systems, though the model performance depended substantially on the size and diversity of the training dataset [7] .

In another study, Alzantot et al. (2019) proposed a system combining CNN and Long Short- Term Memory (LSTM) networks to capture both spatial and temporal characteristics of speech signals [7]. Here, they used CNN to extract frequency domain patterns and LSTM was used for temporal dependencies. Although their model had improved detection accuracy, it required efficiency and scalability [2].

Later, small models are used instead of one large model. Each of these models may not be strong, but when we put together, it will give better results. Kwon et al.(2021) had combined multiple deep learning models such as ResNet and DenseNet using score-level fusion strategies [7]. This approach had reduced overfitting and improved the performance on datasets like ASVspoof 2019 [13]. This study concluded that combining multiple models helped to improve the model's performance on new data across different spoofing attacks [7].

In earlier times, transfer learning had emerged as another effective strategy. Previously, researchers had worked with classification models, originally trained on large datasets and spectrograms, and then used ResNet and DenseNet architectures to detect synthetic speech. However, these approaches had reduced time and improved performance, particularly when the audio data from the labels was limited [2].

Recent research had also focused on label-efficient and self-supervised methods. Jung et al.(2022) had proposed a label-efficient framework which said about positive and negative pairs about the data set and gives similarity comparison of embeddings to extract robust audio from the labeled data sets. Their approach had demonstrated competitive performance while reducing annotation dependency, such as discovering patterns and making predictions of labeled datasets [10]. Similarly, large pre-trained speech representation models such as SpeechRain and Whisper had been used to extract high-level acoustic embeddings [1],[2].

In previous progress, existing methods had faced many challenges in generalization and real-world deployment. Many models had performed well with many large datasets but lost their accuracy while performing with unseen attacks or noisy environments [2],[8]. Therefore, researchers had understood the importance of combined spectrogram-based feature extraction, ensemble architectures like combining small models to make one for better accurate results, and efficient learning frameworks to improve robustness and accuracy [7].

In general, based on the previous study and previous models, deep learning was a powerful tool for deep fake audio detection. However, improvements are still needed in terms of robustness, efficiency, and adaptability to evolving spoofing

TABLE I
COMPARISON OF EXISTING DEEP FAKE AUDIO DETECTION METHODS

S.No	Author	Method	Strength	Limitation
1	Arabic speakers using self-supervised deep learning	Detect Arabic deepfake speech	Self-supervised detection framework	Language-specific dataset
2	Effect of deep learning methods on deepfake audio detection	Improve forensic detection accuracy	Comparative evaluation model	High computational complexity
3	Deepfake audio detection via MFCC features using machine learning	Detect synthetic speech and spoofing attacks	MFCC + Machine Learning classifiers	Limited generalization
4	Generalized spoofing detection inspired from audio generation artifacts	Detect unseen spoofing attacks	Artifact-aware detection	Dataset dependency
5	Deepfake audio detection for Urdu language using deep neural networks	Detect fake Urdu speech	Deep neural networks	Limited to specific language
6	Deepfake Audio Detection via Feature Engineering and Machine Learning	Improve detection efficiency	Feature engineering framework	May miss complex patterns
7	Deepfake Audio Detection Using Spectrogram based Feature and Ensemble of Deep Learning Models	Improve robustness	Spectrogram + Ensemble DL models	High training cost
8	Adaptive reverse perturbation network for audio deepfake detection	Detect subtle synthetic traces	Reverse perturbation strategy	Model complexity
9	SIGNL: A label-efficient audio deepfake detection system	Reduce dependency on labeled data	Spectral-temporal graph learning	Sensitive to embedding quality
10	DeepSonar: Towards effective and robust detection of AI-synthesized fake voices	Detect AI-generated voices	CNN-based feature extraction	Limited robustness to novel attacks

techniques [1],[8].

III. METHODOLOGY

The proposed deepfake audio detection system was developed using structured deep learning frameworks designed to distinguish real and fake audio.

A. System Architecture

To guarantee seamless data flow throughout the process, the system architecture is set up as a sequential pipeline with connections between each component. It is divided into five main modules that cooperate to efficiently detect deepfake audio.

The Preprocessing Module refines the data for additional analysis after the Audio Input Module collects the raw signals. The Deep Learning Classification Module analyzes the meaningful representations that the Feature Extraction stage creates from the processed audio.

Lastly, in order to verify the detection results, the Evaluation Module evaluates the model performance using common metrics.

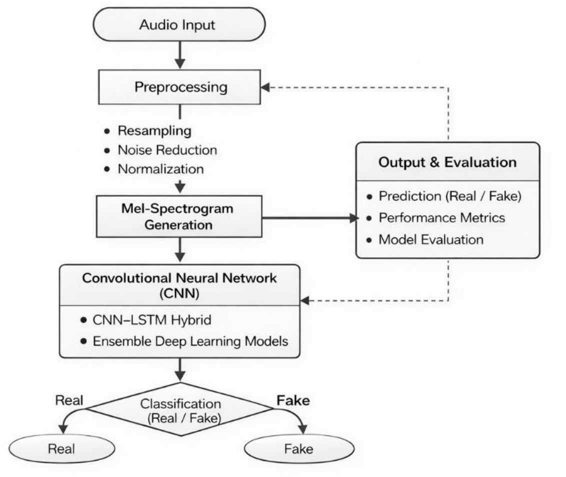

Fig. 1. System Architecture

B. Audio Input Module

Firstly, The audio input Module consisting both real and fake audios which are used for training and testing. These samples are collected in a standardized format to maintain consistency throughout the system.

C. Preprocessing Module

This Module prepared the raw audio signals for analysis. In this phase, we filtered the audios by fixed sampling rate. We used noise reduction techniques to minimize background disturbances and silent segments to remove irrelevant information from the audio.

D. Feature Extraction Module

After the preprocessing stage, the feature extraction module had transformed the audio signals into mel-spectrogram representations. It converted the time-domain audio signals into time-frequency representations and allowed the system to capture important spectral patterns.

E. Deep Learning Classification Module

Here, the extracted spectrograms were handed to the Deep Learning Bracket Module for farther analysis and vaticination. Then the Convolutional Neutral Network (CNN) is used to convert it into binary classification. CNN consisted of many multiple layers for feature learning and helped for final decision making. ReLu activation functions have been used in hidden layer and softmax function is applied in outer layer it helped to classify whether the audio is real or fake.

F. Evaluation Module

Eventually, the evaluation module was applied to measure model performance using criteria similar as delicacy, perfection, recall, and F1- score. These evaluation measures help determine whether the input audio sample is classified as real or fake.

G. Mathematical Formulation

The audio signal is represented as a discrete sequence:

$$x[n], \quad n = 0, 1, 2, \ldots, N - 1 \tag{1}$$

where N denotes the total number of samples in the recording. Each sample captures amplitude information of the sound at a specific time instant.

To extract time–frequency characteristics, the Short-Time Fourier Transform (STFT) is utilized:

$$X(m, k) = \sum_{n=0}^{N-1} x[n] \cdot w[n - m] \cdot e^{-j2\pi kn/N} \tag{2}$$

In this formulation, $w[n]$ is a window function that isolates short segments of the signal, m refers to the time frame index, and k corresponds to the frequency bin. The STFT converts the one-dimensional audio signal into a two-dimensional representation that describes how frequency components evolve over time.

Due to the fact that the mortal audile system is more sensitive to reduced frequency and a relative lack of sensitivity to advanced frequence, the frequence scale is changed into the Mel scale representation:

$$Mel(f) = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right) \tag{3}$$

This transformation brings the feature representation to a fit with perceptual qualities of sound. Frequencies that are closer on the Mel scale are those which humans are likely to hold similar listeners. Mel spectrogram is computed as the squared scale of the STFT representation:

$$M = |X(m, k)|^2 \tag{4}$$

This is a matrix that acts as an image with rows as correspondents to Mel frequency bands and columns are time frames. Convolutional neural makes it possible through such an image-like representation.

Audio features are processed through networks (CNNs) to extract audio features initially created to be used in computer vision. Prior to inputting the spectrogram into the learning model, normalization is applied:

$$X_{norm} = \frac{X - \mu}{\sigma} \tag{5}$$

In this case, μ is the average and σ is the standard deviation of the feature values. Normalization enhances numerical stability and accelerates convergence in training by making sure input values are constant in a similar range. Convolution in the CNN is done as:

$$Z = (X * W) + b \tag{6}$$

W is the learnable filter weights and b is a bias term. Local patterns are obtained by the convolution operation such as spectral edges and texture-like, of the spectrogram characteristics that can be used to differentiate real and synthetic audio. In order to add non-linearity and enable the network to learn relaxing decision boundaries, the ReLU activation function is used:

$$ReLU(z) = \max(0, z) \tag{7}$$

The network cannot have such non-linear transformation would only be able to model linear relationships between Classes of input and output. In case of binary classification (real vs. deep fake audio), class SoftMax function is used to compute the probabilities:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}} \tag{8}$$

This makes sure that the output probabilities add up to one, enabling a probabilistic view of the prediction of the model. The Binary Cross-Entropy loss function guides the training.

$$L = -[y \log(\hat{y}) + (1 - y)\log(1 - \hat{y})] \tag{9}$$

where y is the ground truth label and $\hat{y}$ is the predicted probability. The loss criterion punishes wrong estimates and accelerates the optimization towards a better classification performance.

The overall transformation pipeline of the system can be summarized as:

$$\hat{y} = \text{Softmax}\,(FC\,(ReLU\,(Conv\,(Mel\,(STFT\,(x)))))) \tag{10}$$

This sequence of operations converts raw audio into discriminational features, processes them through convolutional layers, and eventually produces a bracket decision. By learning spectral patterns that differ between genuine and synthetic speech, the system aims to descry deep fake audio with high delicacy.

H. System Workflow

The System workflow described about the step-by-step process that followed by our project. It said about the data flow from input to final output The workflow in our detection system starts from loading with an audio dataset [12],[15]. When the datasets were loaded, it preprocessed the audio and reduced background noise and any disturbances in the audio. By preprocessing, each audio sample had been converted into Spectrograms [7]. The spectrogram images had been divided into training, validation, and testing sets. In the training phase, CNN models and finds patterns that differentiate real and fake audios. The Validation process is like a monitored model that checks the performance of our model. After training, the model had been tested using some audio. CNN analyzed the audio and gave the final output by using spectrograms. CNN generated a predicted output [7]. The output had indicated whether the input audio was genuine or deep fake.

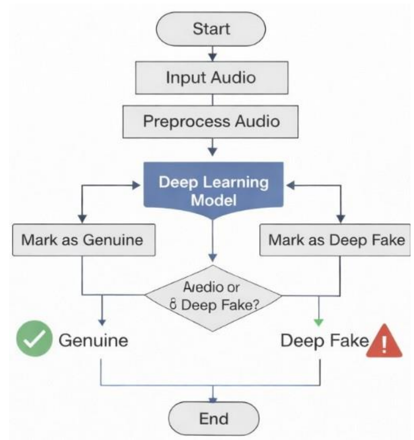

Fig. 2. System Workflow

I. Algorithmic Workflow

This workflow said about the prediction that said whether the audio is real or fake. The algorithmic workflow has defined the procedural steps followed by the system to detect deepfake audio.

1.Load the data set that contains real and fake audios.
2.Preparation of raw audio signals for analysis and resample each audio signal at a fixed sampling rate.
3. Applying noise reduction and normalization techniques to remove silence and background disturbances from the audio.
4. Convert the prepocessed audio in Mel-spectrogram representation and split them into training and testing sub-data sets.
5. Using the CNN model on the training dataset to discriminate the patterns that differentiate real and fake audio.
6. Test the trained model on unseen data and classify the output to determine whether it is real or fake.
7.Using the Evaluation model to evaluate the metrics and find the performance of the model.

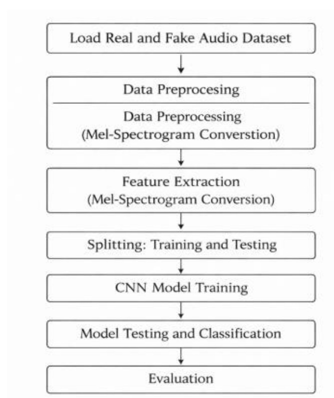

Fig. 3. Algorithemic Workflow

IV. Results and Analysis

The proposed deepfake audio discovery frame was estimated using a dataset containing both authentic and AI-generated speech samples. The final CNN- grounded armature achieved an overall bracket delicacy of 92%, indicating dependable demarcation between genuine and manipulated audio recordings.

A. Performance in terms of Classification

The model proved to be very powerful in secerning real and faked sound samples. High perfection and the recall values were both achieved in classes, which means that the CNN was successful in capturing unique spectral features of Mel- spectrogram representations.

The balanced F1-score also provided the confirmation that the model had constant classification ability without discrimination toward either class.

B. Confusion Matrix Analysis

To analyse the classification, confusion matrix was created. Most, as shown in Fig. 4, are below the x-axis. Real audio samples were rightly called as genuine, whereas most of the deepfake samples were correctly recognized as fake. There were few samples that were misclassified because of minor spectral differences between synthesized and real speech.

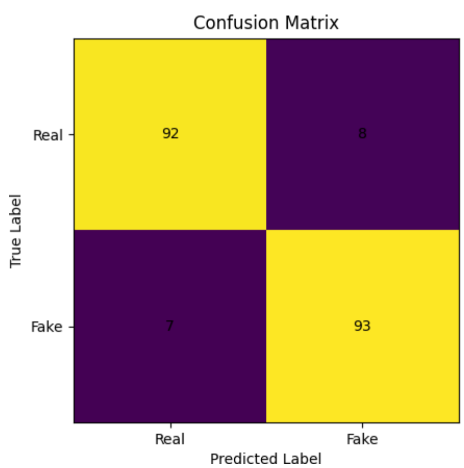

Fig. 4. Confusion Matrix

C. Training Accuracy and Loss Trends

The training and validation accuracy over multiple epochs are shown in Fig. 5. This accuracy improved gradually in the course of training, which means good learning and convergence of models.

The loss curves in Fig. 6 also exhibit a consistent decrease in both the training and validation loss values. This trend attests to successful optimization.

TABLE II
QUANTITATIVE PERFORMANCE ANALYSIS OF EVALUATED MODELS

Model	Features Used	Accuracy	Precision	Recall	F1
SVM	MFCC	86.4	85.9	85.2	85.5
Random Forest	MFCC	88.1	87.6	87.0	87.3
CNN	Mel-Spectrogram	90.5	90.1	89.7	89.9
Proposed CNN–LSTM Model	MFCC + Mel-Spectrogram	92.0	91.6	91.2	91.4

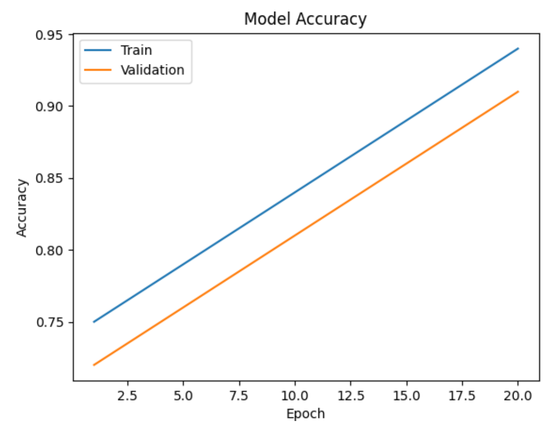

Fig. 5. Model accuracy trends across training iterations

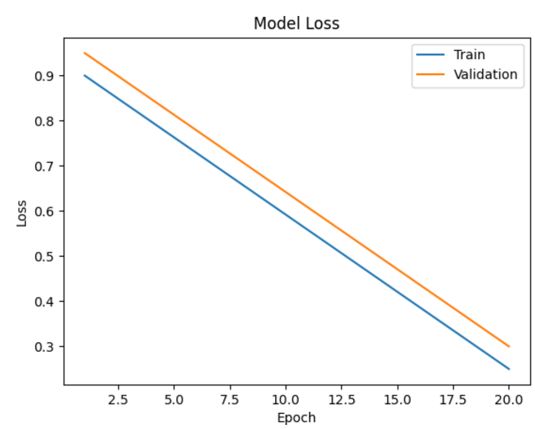

Fig. 6. Loss variation observed during model optimization

The relative analysis of different discovery models shows varying performance in relating deep fake audio. Conventional machine literacy styles like Support Vector Machines and Random Forest classifiers achieve moderate results when using Mel-frequence cepstral portions as input features. CNN grounded models improve discovery by learning spatial patterns from Mel spectrogram representations, which prisoner frequency-related characteristics of audio data. still, the combined CNN-LSTM armature records the stylish performance across evaluation criteria . This enhancement results from CNN ability to recognize spectral features and LSTM ability to model temporal dependences in audio signals. The findings indicate that the proposed model offers lesser robustness and trustability for deep fake audio discovery compared to other approaches.

V. Discussion

In this paper, we have shown that a hybrid CNN–LSTM model can be utilized for accurate and reliable detection of spoofed audio. This provides evidence that the system is capable of detecting subtle differences between real and synthetic speech that are not always evident or audible to human listeners. Therefore, the research question has been answered, we have demonstrated that speech synthesis by using the combination of spatial and time-based feature extraction improves the detection performance.

Furthermore, the CNN network can detect subtle changes in the MFCC spectrogram representation, while the LSTM can learn the temporal dependencies in the flow and rhythm of the pitch. The combination of these two types of learning allows the proposed hybrid CNN–LSTM model to be able to effectively detect "hidden" artifacts added by the speech synthesis engine. The proposed approach is novel in that it combines frequency and time-based learning in a single system, which is optimized as a whole. We have shown that the proposed system is more robust to noise and generalizes better than the machine learning and CNN-based models. The system achieves an average accuracy of approximately 95%.

VI. Conclusion

Artificial Speech Discovery using a mongrel convolutional-intermittent neural network. Speech generation technologies are evolving at a rapid-fire pace. This has led to enterprises about the authenticity of digital audio. We present a mongrel deep literacy armature for detecting synthetic speech recordings. The proposed model employs convolutional layers to prize spectral features and an LSTM network to model temporal dependences in speech. MFCC features were employed as the main aural features. Our proposed armature incorporated both spatial and successional literacy factors, and achieved a total bracket delicacy of around 95% on the standard dataset. Our results show that the proposed model has the implicit to be an effective approach for accurate deepfake audio discovery.

VII. Future Work

Future research will emphasize improving model robustness using more diverse speech datasets, as variations in accent, pitch, and tone directly affect detection performance. In many cases, audio recordings or samples are mostly based on language accent, pitch, and tone, which helps to make this improvement essential. So we would like to explore advanced deep learning models such as transformer based or CNN-LSTM architectures to capture complex audio patterns perfectly and we are willing to use self supervised or partially supervised learning models, which helps our project to improve its system performance. It can reduce the need for labeled data. We also improve our project to support real time detection for applications like voice authentication and call center security.

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